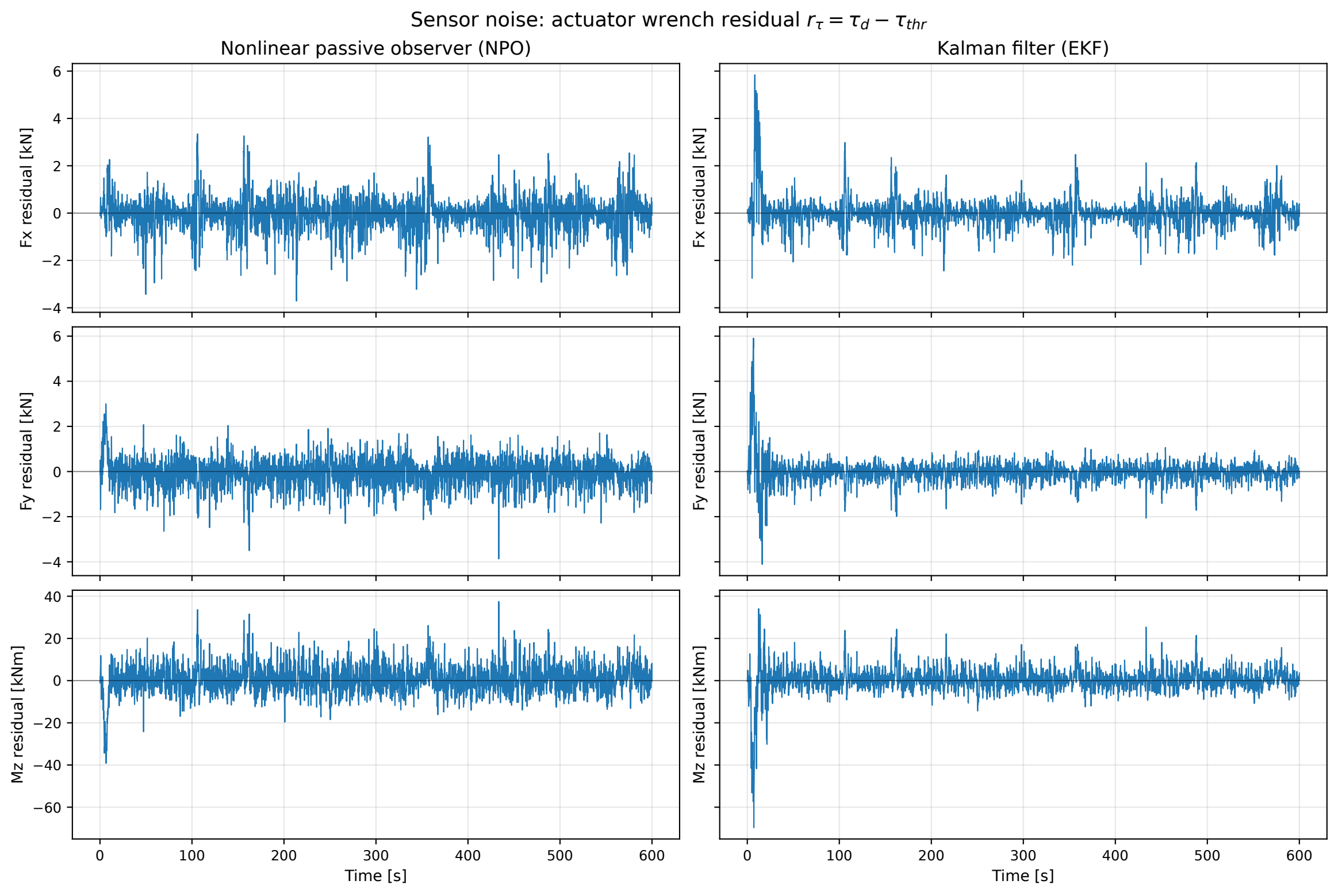

Sensor noise: actuator wrench residual $r_\tau = \tau_d - \tau_{thr}$
Nonlinear passive observer (NPO)
Kalman filter (EKF)
Fx residual [kN]
Fy residual [kN]
Mz residual [kNm]
Time [s]
6
4
2
0
−2
−4
40
20
0
−20
−40
−60
0
100
200
300
400
500
600